REST APIs, Elysia & Kubb

PB138 — Basics of Web Development

"Your frontend and backend finally need to talk. Let's teach them how."

Agenda

1. HTTP — the language of the web
2. Client-Server lifecycle
3. REST API design
4. Building APIs in TypeScript — Express → Elysia + Zod
5. CORS
6. OpenAPI specification
7. React Hooks & TanStack Query
8. Kubb — code generation from specs

HTTP

The protocol that powers the internet — built in 1989, still held together with duct tape and optimism

What is HTTP?

- **H**yper**T**ext **T**ransfer **P**rotocol
- A **request-response** protocol: client asks, server answers
- **Stateless**: every request is independent, server remembers nothing
- Text-based (HTTP/1.1), then binary (HTTP/2, HTTP/3)
- Runs over TCP, usually port `80` (HTTP) or `443` (HTTPS)

```
Client                                Server
│                                          │
│   GET /api/users HTTP/1.1                │
│  ──────────────────────────────────────>│
│                                          │
│   HTTP/1.1 200 OK                        │
│   [{"id":1,"name":"Alice"}]              │
│  <──────────────────────────────────────│
```

HTTP Message Anatomy

```
Request:
GET /api/users?role=admin HTTP/1.1
Host: api.example.com
Authorization: Bearer eyJhbGci...
Content-Type: application/json

{"name": "Alice", "email": "a@b.com"}
```

```
Response:
HTTP/1.1 200 OK
Content-Type: application/json
Location: /api/users/42

{"id": 42, "name": "Alice"}
```

- **Method + Path** — what to do and to which resource
- **Headers** — metadata (auth, content type, caching, CORS...)
- **Body** — data payload (request: POST/PUT/PATCH; response: usually JSON)
- **Query params** — filtering, sorting, pagination (`?role=admin`)
- **Status code** — how it went (in response)

HTTP Methods

Method	Purpose	Has Body?	Idempotent?
`GET`	Read a resource	No	Yes
`POST`	Create a resource	Yes	No
`PUT`	Replace a resource fully	Yes	Yes
`PATCH`	Partially update	Yes	Usually
`DELETE`	Remove a resource	No	Yes

Idempotent = calling it N times gives the same result as calling it once.

`DELETE /users/1` twice → same state. `POST /users` twice → two users.

HTTP Status Codes

Range	Category	Common codes
2xx	Success	`200 OK`, `201 Created`, `204 No Content`
3xx	Redirect	`301 Moved Permanently`, `304 Not Modified`
4xx	Client error	`400 Bad Request`, `401 Unauthorized`, `403 Forbidden`, `404 Not Found`, `409 Conflict`, `422 Unprocessable Entity`
5xx	Server error	`500 Internal Server Error`, `503 Service Unavailable`

> *"Is it a 401 or 403?"*
>
> `401` = who are you? (not logged in)
>
> `403` = I know who you are, you just can't do this

Status Codes in Practice

```
// Correct usage
res.status(201).json(newUser)                // POST — resource created
res.status(200).json(users)                  // GET  — here's your data
res.status(204).send()                       // DELETE — done, nothing to return
res.status(404).json({ error: "User not found" })
res.status(400).json({ error: "Invalid input" })
res.status(401).json({ error: "Login required" })
res.status(403).json({ error: "Admins only" })

// What juniors (and some seniors) do
res.status(200).json({ error: "User not found" })  // wrong
res.status(200).json({ success: false })            // wrong
```

Status codes ARE part of your API contract. Clients branch on them.

Client-Server Lifecycle

What actually happens between `fetch()` *and* `.json()`

The Full Journey

```
React App                                Server
 │                                                │
 │   1. fetch('/api/products')                    │
 │      DNS: resolve hostname → IP                │
 │      TCP: 3-way handshake                      │
 │      TLS: negotiate if HTTPS                   │
 │                                                │
 │   2. HTTP Request ──────────────────────────>  │
 │      GET /api/products                         │   3. Router matches path
 │      Authorization: Bearer xyz                 │   4. Middlewares run (auth, log...)
 │                                                │   5. Route handler executes
 │                                                │   6. Data fetched (DB / memory)
 │   7. HTTP Response <──────────────────────────│
 │      200 OK                                    │
 │      [{"id":1,"name":"Widget"}]                │
 │                                                │
 │   8. React updates state → re-render           │
```

REST API

Constraints you'll hate — until you work with an API that ignores them all

What is REST?

REpresentational **S**tate **T**ransfer — an architectural style, not a protocol.

Defined by Roy Fielding in his **2000 PhD dissertation**. Yes, really.

Core constraints:

- **Stateless** — server stores no session; each request is self-contained
- **Uniform interface** — resources identified by URLs, standard HTTP methods
- **Client-server** — clear separation of concerns
- **Cacheable** — responses indicate whether they can be cached
- **Layered** — client doesn't know if it talks to a proxy or the real server

Resource-Oriented URLs

Think in **nouns**, not verbs.

```
Verb-based (not REST)          Resource-based (REST)

/getUsers                      GET     /api/users
/createUser                    POST    /api/users
/updateUser?id=1               PUT     /api/users/1
/deleteUser?id=1               DELETE  /api/users/1
/getUserPosts?userId=1         GET     /api/users/1/posts
```

The URL identifies **what**.

The HTTP method identifies **what to do with it**.

Richardson Maturity Model

A way to score how "RESTful" an API actually is.

```
Level 3 │  HATEOAS ─── responses include links to next actions
        │
Level 2 │  HTTP Verbs + Status Codes ─── use methods correctly
        │                              ▲ most real-world APIs live here
Level 1 │  Resources ─── separate URL per resource
        │
Level 0 │  Single Endpoint ─── one URL, everything is POST
```

Target: Level 2 — correct HTTP methods + status codes. Browsers and CDNs can cache `GET` automatically. `DELETE` is idempotent. Clients branch on status codes, not body parsing.

Building APIs in TypeScript

From Express to Elysia — how the ecosystem evolved

Express.js

"Node.js framework so minimal, you'll add 40 packages before writing a single route"

What is Express?

- Minimal, unopinionated web framework for Node.js
- Created in 2010, still the most-used Node.js backend framework
- `npm install express` → you have a server

```
import express from 'express'

const app = express()
app.use(express.json()) // parse JSON request bodies

app.get('/api/health', (req, res) => {
  res.json({ status: 'ok' })
})

app.listen(3000, () => {
  console.log('Server on http://localhost:3000')
})
```

That's it. That's a server.

The Middleware Chain

Every request flows through a pipeline of functions.

```
Incoming Request
      |
      v
  express.json()    -- parse request body
      |
      v
  cors()            -- set CORS headers
      |
      v
  authStub()        -- attach currentUser
      |
      v
  Route Handler     -- app.get('/api/users', handler)
      |
      v
  Response sent
```

Middleware — The Anatomy

A middleware is just a function: `(req, res, next) => void`

```
// Logging middleware
const logger = (req, res, next) => {
  console.log(`→ ${req.method} ${req.path}`)
  next() // must call next() or the request hangs forever
}

// Auth stub — Week 8 replaces this with real OAuth
const authMiddleware = (req, res, next) => {
  req.currentUser = { id: '1', role: 'admin' }
  next()
}

app.use(logger)
app.use(authMiddleware)
```

`next()` = "pass control to the next function in the chain."

Forgetting it = request hangs silently. Classic Friday afternoon bug.

Route Handlers

```
const products: Product[] = [] // in-memory for now — Week 5 adds a real DB

// GET all
app.get('/api/products', (req, res) => {
  res.json(products)
})

// GET one
app.get('/api/products/:id', (req, res) => {
    const product = products.find(p => p.id === req.params.id)
    if (!product) return res.status(404).json({ error: 'Not found' })
    res.json(product)
})

// POST — create
app.post('/api/products', async (req, res) => {
  const product = { id: crypto.randomUUID(), ...req.body }
  products.push(product)
  res.status(201).json(product)
})
```

Runtime Validation with Zod

Remember Week 3? `req.body` is `any`. Let's fix that.

```
import { z } from 'zod'

const CreateProductSchema = z.object({
  name: z.string().min(1),
  price: z.number().positive(),
  category: z.enum(['electronics', 'clothing', 'food']),
})

app.post('/api/products', async (req, res) => {
  const result = CreateProductSchema.safeParse(req.body)
  if (!result.success) {
    return res.status(400).json({ error: result.error.flatten() })
  }
  // result.data is fully typed
  const product = { id: crypto.randomUUID(), ...result.data }
  products.push(product)
  res.status(201).json(product)
})
```

Elysia

"Ergonomic framework for humans — end-to-end type safety with Bun"

What is Elysia?

- Modern, ergonomic web framework built for **Bun**
- End-to-end type safety — routes, params, body, response all typed
- Built-in plugin system (CORS, Scalar, etc.)
- Supports **Zod** validation via Standard Schema

```
import { Elysia } from 'elysia'

const app = new Elysia()

app.get('/api/health', () => {
  return { status: 'ok' }
})

app.listen(3000)
```

No `express.json()`, no manual response serialization — just return objects.

Elysia Route Handlers

```
const app = new Elysia()

// GET all
app.get('/api/products', () => {
  return products
})

// GET one
app.get('/api/products/:id', ({ params: { id }, set }) => {
  const product = products.find(p => p.id === id)
  if (!product) {
    set.status = 404
    return { error: 'Not found' }
  }
  return product
})

// POST — create
app.post('/api/products', ({ body }) => {
  const product = { id: crypto.randomUUID(), ...body }
  products.push(product)
  return product
})
```

Same concepts as Express. Cleaner API — destructure what you need, return objects directly.

Elysia + Zod

"One Zod schema to validate, type, and document them all"

The Problem with Separate Concerns

With plain Express, you often maintain things separately:

```
TypeScript interface    →  defines the shape
Zod schema              →  validates the input
Separate tool / config  →  generates the OpenAPI spec
```

Each one can go out of sync. **Elysia + Zod** unifies all three.

Elysia + Zod — Single Source of Truth

One Zod schema = types + validation + OpenAPI spec generation:

```
import { z } from 'zod'

export const CreateProductSchema = z.object({
  name: z.string().min(1).max(100),
  price: z.number().positive(),
  category: z.enum(['electronics', 'clothing', 'food']),
})

// TypeScript type is inferred — no separate interface needed
type CreateProduct = z.infer<typeof CreateProductSchema>
```

Plain Zod — no special wrappers. Elysia validates input and generates OpenAPI docs automatically.

Route Definitions

Routes are defined with chaining — method, path, handler, schema config:

```
import { Elysia } from 'elysia'
import { z } from 'zod'

const app = new Elysia({ prefix: '/products', tags: ['Products'] })

  .get('/', () => {
    return products
  }, {
    detail: { description: 'List of products' },
  })

  .post('/', ({ body }) => {
    return productsService.create(body)
  }, {
    body: CreateProductSchema,
    detail: { description: 'Create a product' },
  })
```

Pass a Zod schema in the third argument — Elysia validates automatically and returns 400 on invalid input

The Evolution — At a Glance

	Express	Express + tsoa	Elysia + Zod
Language	JS / TS	TS	TS
Types	manual	decorators	inferred from Zod
Validation	manual Zod	manual Zod	automatic
OpenAPI	manual / tsoa gen	tsoa gen	built-in
Code gen needed	—	`tsoa generate`	—
Runtime	Node.js	Node.js	Bun

Our assignment uses Elysia + Zod — the rightmost column.

Exercise: Setup & Backend

Getting Started

```
cd pb138-seminars && git checkout seminar-04-assignment
bun install
bun run dev
```

URL	What
http://localhost:3000	REST API
http://localhost:3000/api-docs	Scalar — explore & test all endpoints
http://localhost:5173	React frontend

Each backend module has the same layers:

types → **schema** (Zod) → **repository** (data) → **service** (logic) → **routes** (Elysia)

Read through the `students` **module** first — it's the reference implementation.

Task 1 — Course Filtering

File: `apps/server/src/modules/courses/courses.service.ts`

`getAll()` receives a `filter` object but ignores it. Implement four filters:

Filter	What to do
`filter.semester`	Keep courses matching the given semester
`filter.tags`	Keep courses that have **all** requested tags
`filter.minCredits` / `maxCredits`	Keep courses in credit range (inclusive)
`filter.instructorId`	Keep courses taught by that instructor

Each filter: `if (filter.x) { courses = courses.filter(...) }`

Verify: Scalar → `GET /courses` with query params

Task 2 — Instructor Creation

Step A — Validation (`instructor.schema.ts`)

Add constraints to `CreateInstructorBodySchema`:

- `firstName`, `lastName` — non-empty, max 50 chars
- `email` — valid email format
- `department` — non-empty

Step B — Service (`instructors.service.ts`)

Implement `create()` — pass data to `instructorsRepository.create()`.

Reference: `students` module has the same pattern.

Verify: Scalar → `POST /instructors` — try valid & invalid bodies

CORS

"The browser's way of saying: I don't know you like that"

Same-Origin Policy

Browsers block requests to different origins by default.

Origin = protocol + hostname + port

```
http://localhost:5173     ← your React app (Vite)
http://localhost:3000     ← your Express API

Different ports → different origins → browser blocks it
```

This is a **security feature**, not a bug.

Imagine if any website could call `https://yourbank.com/api/transfer` using your logged-in session cookies... yeah.

CORS — The Solution

Cross-**O**rigin **R**esource **S**haring — the server tells the browser which origins it trusts.

```
// Express
import cors from 'cors'
app.use(cors({ origin: ['http://localhost:5173'] }))

// Elysia — plugin
import { cors } from '@elysiajs/cors'
app.use(cors({ origin: ['http://localhost:5173'] }))
```

The server adds `Access-Control-Allow-Origin` headers.
The browser checks them before showing you the response.

For non-simple requests (custom headers like `Authorization`), the browser sends an `OPTIONS` preflight first. If the server doesn't respond with the right CORS headers → the famous CORS error.

Works fine in Postman but breaks in the browser? It's always CORS.

Checkpoint: CORS

- Same-Origin Policy: browsers block cross-origin requests by default
- CORS lets your server declare which origins it trusts
- Never use `origin: '*'` in production (unless it's a truly public API)
- Set `credentials: true` if using cookies or Authorization headers
- CORS errors = server didn't include the right response headers

Questions?

OpenAPI

"Because 'just read the source code' is not documentation"

What is OpenAPI?

A standard, machine-readable description of your HTTP API.

One spec, many superpowers:

- Interactive docs (Scalar, Swagger UI, Redoc)
- Typed client generation — Kubb, orval, openapi-typescript
- Request/response validation
- Contract between frontend and backend teams
- Mock servers for testing before the backend exists

Previously known as **Swagger**. OpenAPI 3.x is the current standard.

Schema-First vs Code-First

Two approaches to owning the spec:

```
Schema-First                    Code-First
────────────────                ────────────────
Write openapi.yaml              Write routes with Zod schemas
│                               │
▼                               ▼
Generate server stubs           Elysia + @elysiajs/openapi extracts the spec
Generate client code            │
│                               ▼
▼                               Generate openapi.json
Implement the handlers
```

We use code-first — write your routes with Zod schemas, generate the spec.

More natural, stays in sync with your code automatically.

No YAML authoring by hand at 2am.

OpenAPI Spec — A Peek

```yaml
openapi: 3.0.0
info:
  title: Products API
  version: 1.0.0
paths:
  /api/products:
    get:
      summary: List all products
      responses:
        '200':
          content:
            application/json:
              schema:
                type: array
                items:
                  $ref: '#/components/schemas/Product'
components:
  schemas:
    Product:
      type: object
      required: [id, name, price, category]
      properties:
        id:    { type: string }
        name:  { type: string }
        price: { type: number }
        category: { type: string, enum: [electronics, clothing, food] }
```

Scalar — Your API Playground

Reads `openapi.json` and gives you interactive docs with modern DX:

- Cleaner design, dark mode, auto-generated code examples
- Built-in request builder with history
- Sends real requests, shows live responses
- Every endpoint browsable — no source code reading needed

Alternative: **Swagger UI** — the classic, still widely used. Same idea, older UI.

We use **Scalar** on this course — available at `/api-docs` when you run the server.

Checkpoint: OpenAPI

- Machine-readable API description — one source of truth
- Enables docs, client generation, validation, mock servers
- Schema-first: write YAML → generate code
- Code-first: write code → generate YAML
- We use code-first — stays in sync automatically
- Scalar = interactive docs + live API playground

Questions?

React Hooks

The functions that make React components actually do things

Built-in Hooks — The Essentials

```
// useState — reactive state
const [count, setCount] = useState(0)
const [user, setUser] = useState<User | null>(null)

// useEffect — side effects (API calls, subscriptions, timers)
useEffect(() => {
  console.log('Component mounted')
  return () => console.log('Component unmounted') // cleanup
}, []) // [] = run once on mount

useEffect(() => {
  console.log('Count changed to', count)
}, [count]) // runs when count changes

// useMemo — expensive computations, cached until deps change
const sorted = useMemo(() => users.sort(byName), [users])

// useRef — mutable value that doesn't trigger re-render
const inputRef = useRef<HTMLInputElement>(null)
```

Rules: only call hooks at the **top level** — never inside `if`, loops, or callbacks.

Custom Hooks

Extract reusable logic into functions starting with `use`.

```
// useLocalStorage — persist state across page reloads
function useLocalStorage<T>(key: string, initial: T) {
  const [value, setValue] = useState<T>(() => {
    const stored = localStorage.getItem(key)
    return stored ? JSON.parse(stored) : initial
  })

  useEffect(() => {
    localStorage.setItem(key, JSON.stringify(value))
  }, [key, value])

  return [value, setValue] as const
}

// Usage — same API as useState, but survives refresh
const [theme, setTheme] = useLocalStorage('theme', 'light')
const [cart, setCart] = useLocalStorage<CartItem[]>('cart', [])
```

Custom hooks = composable, testable, reusable pieces of stateful logic.

TanStack Query — Data Fetching Done Right

```
import { useQuery, useMutation, useQueryClient } from '@tanstack/react-query'

function ProductList() {
  const { data, isLoading, error } = useQuery({
    queryKey: ['products'],
    queryFn: () => fetch('/api/products').then(r => r.json()),
  })

  if (isLoading) return <p>Loading...</p>
  if (error) return <p>Error: {error.message}</p>
  return <ul>{data.map(p => <li key={p.id}>{p.name}</li>)}</ul>
}
```

What you get **for free**: caching, background refetching, deduplication, retry on error, loading/error states, window focus refetching, stale-while-revalidate.

`queryKey` = cache key. Same key → same cached data across components.

`queryFn` = how to fetch. TanStack Query handles **when** and **how often**.

Kubb

Stop writing `fetch('/api/users')` *by hand like it's 2015*

Kubb — Code Generation from OpenAPI

```
Elysia Routes -->  openapi.json  -->  Generated Client
                  (kubb.config.ts)

GET    /products                        getProducts()
POST   /products                        createProduct(data)
GET    /products/:id                    getProductById(id)
PUT    /products/:id                    updateProduct(id, data)
DELETE /products/:id                    deleteProduct(id)
```

All fully typed. No manual fetch. Backend changes → TypeScript yells at you.

What Kubb Generates

Kubb generates **query options factories** — functions returning `{ queryKey, queryFn }`:

```
// Generated by Kubb — you never write this
export function getProductsQueryOptions() {
  return {
    queryKey: ['products'],
    queryFn: () => fetch('/api/products').then(r => r.json()) as Promise<Product[]>,
  }
}
```

You use them with standard `useQuery` — no magic hooks, full control:

```
import { useQuery } from '@tanstack/react-query'
import { getProductsQueryOptions } from '@/api/generated'

// Basic usage — just spread the generated options
const { data, isLoading } = useQuery(getProductsQueryOptions())

// Override anything — e.g. poll every 5s for a live dashboard
const { data } = useQuery({
  ...getProductsQueryOptions(),
  refetchInterval: 5000,
})

// Disable the query until user picks a category
const { data } = useQuery({
  ...getProductsByCategory(selectedCategory),
  enabled: selectedCategory !== null,
})
```

Same `useQuery` you already know. Kubb generates the boring parts — you keep full control.

Why Kubb

- **Type safety end-to-end** — backend types flow to the frontend automatically
- **No drift** — spec changes → regenerate → TypeScript catches all breakages
- **Zero manual fetch code** — routes, types, React Query hooks all generated
- **Supports many outputs** — Axios, fetch, React Query, SWR, Zod validators

```
# Add to package.json
"generate": "kubb generate"

# Run whenever the backend changes:
npm run generate
```

The OpenAPI spec becomes the **single source of truth**.

Frontend and backend can evolve independently — safely.

Exercise: Frontend

Task 3a — Student List

File: `apps/web/src/pages/StudentsPage.tsx`

Display all students using the generated query options:

```
import { useQuery } from '@tanstack/react-query'
import { getStudentsQueryOptions } from '@/generated'

const { data, isLoading, error } = useQuery(getStudentsQueryOptions())
```

Handle loading/error states. Render each student's name, email, and UCO.

Reference: `CoursesPage.tsx` — same pattern, different entity.

Task 3b — Semester Filter

File: `apps/web/src/pages/CoursesPage.tsx`

Add a semester filter to the existing courses page:

1. Add state: `const [semester, setSemester] = useState<string>()`
2. Pass to query: `getCoursesQueryOptions({ semester })`
3. Render a `<Select>` above the course grid: All / Spring / Fall

Verify: http://localhost:5173 — dropdown should filter courses live

Summary

What We Covered

1. **HTTP** — message anatomy, methods, status codes
2. **Client-Server lifecycle** — DNS → TCP → middleware → handler → response
3. **REST** — stateless, resource-oriented, target Level 2
4. **Express.js** — middleware chain, route handlers, Zod validation
5. **Elysia** — modern, ergonomic, built for Bun
6. **Elysia + Zod** — single source of truth: types + validation + OpenAPI
7. **CORS** — why it exists, how to configure it correctly
8. **OpenAPI** — machine-readable spec, code-first approach
9. **React Hooks & TanStack Query** — useQuery, caching, background refetching
10. **Kubb** — typed query options generated from the OpenAPI spec

"A good API is not built once. It's maintained with your future self in mind."

Now let's build something.

Exercises time!